

Question 1: House Price Prediction using Linear Regression

Aim

To develop a House Price Prediction using the Linear Regression algorithm

Tools Required

- Python
- Pandas
- Scikit-learn (sklearn)
- VS Code / Jupyter Notebook

Code

```
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

data = fetch_california_housing()
X = data.data
y = data.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("R2 Score:", round(r2_score(y_test, y_pred), 2))
```

Output

```
> house price prediction.py > ...
1  from sklearn.datasets import fetch_california_housing
2  from sklearn.model_selection import train_test_split
3  from sklearn.linear_model import LinearRegression
4  from sklearn.metrics import r2_score
5  data = fetch_california_housing()
6  X = data.data
7  y = data.target
8  X_train, X_test, y_train, y_test = train_test_split(
9      |   X, y, test_size=0.2, random_state=42
10 )
11 model = LinearRegression()
12 model.fit(X_train, y_train)
13 y_pred = model.predict(X_test)
14 print("R2 Score:", round(r2_score(y_test, y_pred), 2))
```

Ports

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\chandravardhan reddy\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\chandravardhan reddy\AppData\Local\Programs\Microsoft VS Code\Code.exe" "e:/house price prediction.py"
R2 Score: 0.58
PS C:\Users\chandravardhan reddy\AppData\Local\Programs\Microsoft VS Code> 
```

Result

House Price Prediction using the Linear Regression algorithm successfully implemented